

SAVITHA VIJAYARANGAN

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EXPERIENCE

IDX Exchange · *NLP Engineer Intern* Remote, US · Sep 2026 – Present

- Delivered sub-100ms semantic search over 10,000+ listings with sentence-transformer embeddings and FAISS, evaluated against BM25 and served through an 8-endpoint FastAPI service with caching and rate limiting.
- Reached 87% F1 on information extraction and 92% query-parsing accuracy with a named-entity extractor and an object-oriented natural-language query parser that generates parameterized SQL with schema validation.
- Built a Fair Housing compliance checker with 100% recall and 84% precision on labeled violations, and a 200+ term real-estate taxonomy from n-gram analysis of agent listing remarks.

San José State University · *Instructional Student Assistant, Deep Learning (DATA 255) & Generative AI (DATA 266)* San Jose, CA · Jul 2026 – Present

- Developed and refined course curriculum and mentored 100+ graduate students on model deployment and distributed training.
- Automated grading pipelines and submission-management scripts for high-enrollment graduate courses.

HexasenseAI · *ML Engineer Intern* Austin, TX (Remote) · Jun 2025 – Aug 2025

- Deployed optimized ML pipelines that reduced inference latency by 30%, improving real-time system responsiveness.
- Built clinical risk-prediction models on healthcare data, applying Responsible AI practices for fairness, transparency and reliability.

Larsen & Toubro Infotech (LTI) · *System Consultant (Data & Analytics Engineer); Software Engineer* India · Sep 2017 – May 2021

- Built Python, SQL and Airflow ETL pipelines handling 1M+ records/day, integrated 5+ sources into a Snowflake warehouse, and cut pipeline latency by 25%.
- Built Python backend services handling 50K+ transactions/day and delivered 10+ production features via REST APIs in Agile sprints.

PROJECTS

Wildfire Edge Sentinel · HP Edge AI SJSU Hackathon · YOLO11, Qwen2.5-VL, LoRA, ONNX, Docker Sep 2026

- Built offline-first wildfire smoke detection on an HP ZGX Nano: a fine-tuned YOLO11 detector plus a Qwen2.5-VL-7B LoRA distilled from a 32B teacher; the pipeline keeps detecting and queues alerts during network outages.
- Raised detector mAP50 from 0.002 to 0.748 on D-Fire and from 0.198 to 0.718 on lookout-tower smoke; raised VLM source-type agreement from 45.0% to 73.6% on 500 held-out crops.
- Designed edge-vs-cloud escalation that keeps video on-device: on a 266-frame, 6-tower run, cut VLM calls from 266 to 11, tokens from 447K to 3K and video upload from 23 MB to 0 B.

Kayak Travel Simulation System · FastAPI, Kafka, LangChain, OpenAI API, MySQL, MongoDB, Redis, React

- Engineered 3-tier microservices with FastAPI and Kafka for real-time booking feeds, on a MySQL, MongoDB and Redis data layer deployed with Docker.
- Built an AI chatbot with LangChain and the OpenAI API to identify user intent, improving response accuracy by 25%.

GPT-Style Language Model from Scratch · PyTorch, Transformers, Attention Feb – Mar 2026

- Built and trained a ~4.8M-parameter decoder-only Transformer from scratch (no pretrained embeddings) on TinyStories, reaching 0.6438 validation loss; documented failure modes across sampling temperatures.

Image Classification & Super-Resolution on NPU · PyTorch, CNN, ONNX, NPU deployment

- Trained a custom CNN for multi-class ImageNet classification and a super-resolution DNN, and deployed both for real-time inference on NPU hardware by converting PyTorch models to ONNX and compiling to MXQ.

EDUCATION

San José State University · M.S., Applied Data Intelligence · GPA 3.53 · Coursework: ML, Deep Learning, Generative AI, Distributed Systems, Big Data 2025 – Present

B.S. Abdur Rahman University · B.Tech., Electronics & Communication Engineering 2013 – 2017

Harvard Business School Online · Business Analytics Certification 2023

SKILLS

ML / DL: PyTorch, Scikit-Learn, TensorFlow (coursework), ONNX, CNNs, Transformers

GenAI / NLP: LLMs, RAG, LangChain, OpenAI API, LoRA/QLoRA, vector databases, FAISS, sentence-transformers, NLTK

Deployment & data: FastAPI, Docker, MLflow, ONNX/NPU deployment, AWS S3, PySpark, Kafka, Airflow, dbt, Snowflake

Languages & databases: Python, SQL, PostgreSQL, MySQL, MongoDB, Redis